

Term	Definition
Ability	A term used in human resource management denoting an acquired or natural capacity or talent that enables an individual to perform a particular task successfully. (INCOSE SECF)
Acquirer	The stakeholder that acquires or procures a product or service from a supplier. (ISO/IEC/IEEE 15288, 2023)
Activity	A set of cohesive tasks of a process. (ISO/IEC/IEEE 15288, 2023)
Agile systems-engineering	An SE process using agile approach.
Agile-systems engineering	An engineering process producing agile systems.
Agreement	The mutual acknowledgment of terms and conditions under which a working relationship is conducted. (ISO/IEC/IEEE 15288, 2023)
Architect	See System architect.
Architecture	See System architecture.
Artifact	Work product that is produced and used during a project to capture and convey information. (ISO/IEC/IEEE 15288, 2023)
Attribute	An attribute of a system (or system element) is an observable characteristic or property of the system

Term	Definition
	(or system element).
Baseline	An agreed-to description of the attributes of a product at a point in time, which serves as a basis for defining change. (EIA-649C, 2019)
Behavior	The way in which one acts or conducts oneself, especially towards others. (INCOSE SECF)
Black box	Black box represents an external view of the system (attributes). Also referred to as opaque box.
Brownfield SE	Development of “to-be” system or system elements in the presence of existing or legacy “as-is” system or system elements. Note: A brownfield approach is usually used to extend, improve, or replace a system that is in use or to reuse system elements that will not be impacted by the desired changes. The new system architecture must take into account the existing system elements and functions, which impose constraints on the overall system definition.
Capability	An expression of a system, product, function, or process ability to achieve a specific objective under stated conditions.
Commonality	(Of a product line) refers to functional and non-functional characteristics that can be shared with all member products within a product line. (ISO/IEC/IEEE 26550, 2015)

Term	Definition
Competence	The measure of specified ability to do something well. (INCOSE SECF)
Competency	An observable, measurable set of skills, knowledge, abilities, behaviors, and other characteristics an individual needs to successfully perform work roles or occupational functions. Competencies are typically required at different levels of proficiency depending on the specific work role or occupational function. Competencies can help ensure individual and team performance aligns with the organization's mission and strategic direction. (INCOSE SECF)
Configuration item (CI)	A system, system element, or artifact designated for configuration management.
Customer	See Acquirer.
Decision gate	A decision gate is an approval event (may be associated with a review). Entry and exit criteria are established for each decision gate; continuation beyond the decision gate is contingent on the agreement of decision makers.
Design constraints	The boundary conditions, externally or internally imposed, for the SoI within which the organization must remain when executing the processes during the concept and development stages.
Engineered System	A system designed or adapted to interact with an anticipated operational environment to achieve

Term	Definition
	one or more intended purposes while complying with applicable constraints. (INCOSE Definitions, 2019)
Enterprise	A purposeful combination of interdependent resources that interact with each other to achieve business and operational goals. (Rebovich and White, 2011)
Environment	The surroundings (natural or man-made) in which the SoI is utilized and supported or in which the system is being developed, produced, and retired.
Facility	The physical means or equipment for facilitating the performance of an action, for example, buildings, instruments, and tools.
Failure	The event in which any part of a system or system element does not perform as required by its specification. Note: The failure may occur at a value in excess of the minimum required in the specification, that is, past design limits or beyond the margin of safety.
Functional configuration audit (FCA)	An evaluation to ensure that the product meets baseline functional and performance capabilities. (Adapted from ISO/IEC/IEEE 15288, 2023)
Greenfield SE	Development of a system for a new environment and set of user scenarios and requirements. Note: A greenfield approach typically has no significant legacy constraints or dependencies within the system boundary. However, it is rare that there are

Term	Definition
	no constraints or dependencies from external interfaces or enabling systems.
Human factors	The systematic application of relevant information about human abilities, characteristics, behavior, motivation, and performance.
Interface	A shared boundary between two systems or system elements, defined by functional characteristics, common physical interconnection characteristics, signal characteristics, or other characteristics, as appropriate. (Adapted from ISO/IEC 2382, 2015)
IPO diagram	Figures in this handbook that provide a high-level view of the process of interest. The diagram summarizes the process activities and their typical inputs and typical outputs from/to other processes or external actors.
Knowledge	A body of information applied directly to the performance of a function. (INCOSE SECF)
Life cycle cost (LCC)	The total cost of a system over its entire life. Note: It includes all costs associated with the system and its use in the concept, development, production, utilization, support, and retirement stages.
Life cycle model	A framework of processes and activities concerned with the life cycle, which also acts as a common reference for communication and understanding. (ISO/IEC/IEEE 15288, 2023)

Term	Definition
Measure	Variable to which a value is assigned as the result of measurement. (ISO/IEC/IEEE 15939, 2017)
Measurement	Set of operations having the object of determining a value of a measure. (ISO/IEC/IEEE 15939, 2017)
Measures of effectiveness (MOEs)	Measures that define the acquirer's key indicators of achieving the mission needs for performance, suitability, and affordability across the life cycle.
Measures of performance (MOPs)	Measures to assess whether the system meets design or performance requirements and has the capability to achieve operational objectives.
N ² diagrams	Graphical representation used to define the internal operational relationships or external interfaces of the SoI.
Need statement	The result of a formal transformation of one or more life cycle concepts into an agreed-to expectation for an entity to perform some function or possess some quality. (INCOSE GtWR, 2022)
Operator	See User.
Organization	Person, or a group of people, and facilities with an arrangement of responsibilities, authorities, and relationships. (Adapted from ISO 9001, 2015)
Performance	A quantitative measure characterizing a physical or functional attribute relating to the execution of a process, function, activity, or task; performance

Term	Definition
	attributes include quantity (how many or how much), quality (how well), timeliness (how responsive, how frequent), and readiness (when, under which circumstances).
Physical configuration audit (PCA)	An evaluation to ensure that the operational system conforms to the operational and configuration documentation. (Adapted from ISO/IEC/IEEE 15288, 2023)
Process	A set of interrelated or interacting activities that transforms inputs into outputs. (Adapted from ISO 9001, 2015)
Product line	Group of products or services sharing a common, managed set of features that satisfy specific needs of a selected market or mission. (ISO/IEC/IEEE 24765, 2017)
Project	An endeavor with defined start and finish criteria undertaken to create a product or service in accordance with specified resources and requirements. (ISO/IEC/IEEE 15288, 2023)
Proof of concept	A realization of an idea or technology to demonstrate its feasibility.
Quality Characteristics	Inherent characteristic of a product, process, or system related to a requirement. (ISO/IEC/IEEE 15288, 2023)

Term	Definition
Requirement statement	The result of a formal transformation of one or more needs or parent requirements into an agreed-to obligation for an entity to perform some function or possess some quality. (INCOSE GtWR, 2022)
Resource	An asset that is utilized or consumed during the execution of a process. (ISO/IEC/IEEE 15288, 2023)
Return on investment	Ratio of revenue from output (product or service) to development and production costs, which determines whether an organization benefits from performing an action to produce something. (ISO/IEC/IEEE 24765, 2017)
Reuse	The use of an asset in the solution of different problems. (IEEE 1517, 2010)
Skills	An observable competence to perform a learned psychomotor act.
Stage	A period within the life cycle of an entity that relates to the state of its description or realization. Note: Typical life cycle stages include concept, development, production, utilization, support, and retirement.
Stakeholder	A party having a right, share, or claim in a system or in its possession of characteristics that meet that party's needs and expectations.

Term	Definition
Supplier	An organization or an individual that enters into an agreement with an acquirer for the supply of a product or service. (ISO/IEC/IEEE 15288, 2023)
System	An arrangement of parts or elements that together exhibit behavior or meaning that the individual constituents do not. (INCOSE Definitions, 2019)
System architect	The person, team, or organization responsible for a system's architecture, for coordinating engineering effort towards devising solutions to complex problems, and overseeing their implementations.
System architecture	The fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity and its related life cycle processes. (ISO/IEC/IEEE 42020, 2019)
System element	Member of a set of elements that constitutes a system. (ISO/IEC/IEEE 15288, 2023)
System life cycle	The evolution with time of a SoI from conception to retirement.
System of interest (SoI)	The system whose life cycle is under consideration. (ISO/IEC/IEEE 15288, 2023)
System of systems	A SoI whose system elements are themselves systems; typically, these entail large-scale interdisciplinary problems with multiple, heterogeneous, distributed systems.

Term	Definition
Systems engineering	A transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods. (INCOSE Definitions, 2019)
Tailoring	The manner in which any selected issue is addressed in a particular project. Tailoring may be applied to various aspects of the project, including project documentation, processes, and activities performed in each life cycle stage, the time and scope of reviews, analysis, and decision making consistent with all applicable statutory requirements.
Technical performance measures (TPMs)	Measures to assess design progress, compliance to performance requirements, or technical risks and provide visibility into the status of important project technical parameters to enable effective management, thus enhancing the likelihood of achieving the technical objectives of the project.
Trade-off	Decision-making actions that selects from various alternatives on the basis of net benefit to the stakeholders.
User	An individual who, or an organization that, contributes to the functionality of a system and draws on knowledge, skills, and procedures to contribute to the function. Individual who or group that benefits from a system during its utilization.

Term	Definition
Validation	Confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled. (ISO/IEC/IEEE 15288, 2023)
Value	A measure of worth (e.g., benefit divided by cost) of a specific product or service by a customer, and potentially other stakeholders. (McManus, 2005)
Variability	(Of a product line) refers to characteristics that may differ among members of the product line. (ISO/IEC/IEEE 26550, 2015)
Verification	Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled. (ISO/IEC/IEEE 15288, 2023)
Waste	Work that adds no value to the product or service in the eyes of the customer. (Womack and Jones, 1996)
White box	White box represents an internal view of the system (attributes and structure of the elements). Also referred to as transparent box.

APPENDIX D: N² DIAGRAM OF SYSTEMS ENGINEERING PROCESSES

Note: [Figure D.1](#) in this appendix provides an N^2 diagram (see [Section 3.2.4](#)) of the typical inputs and outputs that appear in the IPO diagrams in this handbook. The off-diagonal squares represent the typical inputs/outputs shared by the processes that intersect at a given square. Outputs flow horizontally, inputs flow vertically, and the diagram can be read in a clockwise fashion. These typical inputs and outputs represent “a” way that the SE processes can be performed, but not necessarily “the” way that they must be performed. The absence of a relationship between any two processes does not preclude tailoring to create a relationship. Definitions of the typical inputs and outputs on the IPO diagrams can be found in Appendix E.

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FIGURE D.1 Input/output relationships between the various SE processes. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices pages. All other rights reserved.

The system life cycle processes are placed on the diagonal, abbreviated as follows:

Abbreviation	Life Cycle Process	Handbook Section
ACQ	Acquisition	2.3.2.1
SUP	Supply	2.3.2.2
LCMM	Life Cycle Model Management	2.3.3.1
INFRAM	Infrastructure Management	2.3.3.2
PM	Portfolio Management	2.3.3.3
HRM	Human Resource Management	2.3.3.4
QM	Quality Management	2.3.3.5
KM	Knowledge Management	2.3.3.6
PP	Project Planning	2.3.4.1
PAC	Project Assessment and Control	2.3.4.2
DM	Decision Management	2.3.4.3
RM	Risk Management	2.3.4.4
CM	Configuration Management	2.3.4.5
INFOM	Information Management	2.3.4.6
MEAS	Measurement	2.3.4.7
QA	Quality Assurance	2.3.4.8

Abbreviation	Life Cycle Process	Handbook Section
BMA	Business or Mission Analysis	2.3.5.1
SNRD	Stakeholder Needs and Requirements Definition	2.3.5.2
SRD	System Requirements Definition	2.3.5.3
SAD	System Architecture Definition	2.3.5.4
DD	Design Definition	2.3.5.5
SA	System Analysis	2.3.5.6
IMPL	Implementation	2.3.5.7
INT	Integration	2.3.5.8
VER	Verification	2.3.5.9
TRAN	Transition	2.3.5.10
VAL	Validation	2.3.5.11
OPER	Operation	2.3.5.12
MAINT	Maintenance	2.3.5.13
DISP	Disposal	2.3.5.14
TLR	Tailoring	4.1

In addition to the individual system life cycle processes, the following are also placed on the diagonal, abbreviated as follows:

Abbreviation	Name	Description
EXT	External	External represents those typical inputs and outputs that come from, or go to, beyond the set of system life cycle processes (i.e., they do not come from, or go to, another system life cycle process). Note that these can be either internal (e.g., Organization strategic plan) or external (e.g., Applicable laws and regulations) to the organization.
CTL	Controls	Controls represent those typical inputs and outputs that control, or limit, the execution of the system life cycle processes. They either come in as an external (EXT) typical input or from one or more life cycle processes. They go into every system life cycle process and are shown in Figure 2.11 .
ENAB	Enablers	Enablers represent those typical inputs and outputs that enable, or assist in, the execution of the system life cycle processes. They either come in as an external (EXT) typical input or from one or more life cycle processes. They go into every system life cycle process and are shown in Figure 2.11 .
SIT	Situational	Situational represents those typical inputs and outputs that are situational with respect to the execution of the system life cycle processes (i.e., they

Abbreviation**Name****Description**

are invoked when needed). They can come from any life cycle process. They go into a select number of system life cycle processes, specifically: Decision Management, Risk Management, Configuration Management, Information Management, and System Analysis.

APPENDIX E: INPUT/OUTPUT DESCRIPTIONS

Note: This appendix is a alphabetical list of all the typical inputs and outputs that appear in the IPO diagrams in this handbook. Sources of descriptions are as indicated. These typical inputs and outputs represent “a” way that the SE processes can be performed, but not necessarily “the” way that they must be performed. Other related terms and definitions can be found in Appendix B.

Typical Input/Output	Description
Accepted system or system element	System or system element (product or service) accepted by an acquirer from a supplier consistent with the delivery conditions of the supply agreement.
Acquired system or system element	System or system element (product or service) delivered to the acquirer from a supplier consistent with the delivery conditions of the acquisition agreement.
Acquisition agreement	Mutual acknowledgment of terms and conditions under which a working relationship is conducted between an acquirer and a supplier. (Adapted from ISO/IEC/IEEE 15288, 2023).
Acquisition need	Identified need that cannot be met within the organization encountering the need or a need that can be met in a more economical way by a supplier.
Acquisition payment	Payments or other compensations for an acquired system.
Acquisition records/artifacts	Permanent, readable form of data, information, or knowledge related to acquisition.
Acquisition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the acquisition activities.

Typical Input/Output	Description
Acquisition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform acquisition.
Agreements	Agreements from all applicable life cycle processes, including: acquisition agreement and supply agreement.
Alternative solution classes	Identifies and describes the classes of solutions that may address the problem or opportunity.
Analysis situations	Analyses that arise from any stakeholder. Can originate from any life cycle process.
Applicable laws and regulations	International, national, or local laws or regulations.
Breakdown structures	Hierarchical representations of project aspects into smaller components providing the necessary frameworks to accomplish the project objectives and create the required deliverables.
Business or mission analysis records/artifacts	Permanent, readable form of data, information, or knowledge related to business or mission analysis.
Business or mission analysis report	An account prepared for interested parties in order to communicate the

Typical Input/Output	Description
	status, results, and outcomes of the business or mission analysis activities.
Business or mission analysis strategy/approach	Approaches, schedules, resources, and specific considerations required to perform business or mission analysis.
Business plan	The overall organization business plan, including the business objectives.
Candidate items for configuration management	Items for configuration control. Can originate from any life cycle process.
Candidate items for information management	Items for information control. Can originate from any life cycle process.
Candidate risks and opportunities	Risks and opportunities that arise from any stakeholder. Can originate from any life cycle process.
Change request	Identified anomaly, required, or recommended enhancement to a project, from the time an idea is recorded until the disposition by a designated change authority. (Adapted from ISO/IEC/IEEE 24765, 2017).
Concept of operations (ConOps)	At the organization level, addresses the leadership's intended way of operating the organization (ISO/IEC/IEEE 29148, 2018).

Typical Input/Output	Description
Configuration baseline	Configuration information formally designated at a specific time during the life of a product, product component, service, or service component. (Adapted from ISO/IEC/IEEE 24765, 2017).
Configuration management records/artifacts	Permanent, readable form of data, information, or knowledge related to configuration management.
Configuration management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the configuration management activities.
Configuration management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform configuration management.
Configuration management system	System used to support and enable configuration management.
Configuration verification and audit report	Provides results of configuration management verifications and audits to ensure adequate traceability, control, and visibility. It includes evaluation criteria.
Constraints on solution	Externally imposed limitation on the system, its design, or implementation or on the process used to develop or modify a system. (ISO/IEC/IEEE 29148, 2018).

Typical Input/Output	Description
Critical performance measurement data	Data provided for the identified system-of-interest measurement needs.
Critical performance measurement needs	Identified information needs of the decision makers with respect to system-of-interest expectations.
Customer satisfaction inputs	Responses to customer satisfaction surveys or other instruments.
Decision management records/artifacts	Permanent, readable form of data, information, or knowledge related to decision management.
Decision management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the decision management activities.
Decision management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform decision management.
Decision register	A repository that supports the availability for use and communication of all relevant decision information in a timely, complete, valid, and, if required, confidential manner.
Decision situations	Decisions that arise from any stakeholder. Can originate from any life cycle process.

Typical Input/Output	Description
Design definition records/artifacts	Permanent, readable form of data, information, or knowledge related to design definition.
Design definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the design definition activities.
Design definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform design definition.
Disposal procedure	Presents an ordered series of steps to perform disposal.
Disposal records/artifacts	Permanent, readable form of data, information, or knowledge related to disposal.
Disposal report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the disposal activities.
Disposal strategy/approach	Approaches, schedules, resources, and specific considerations required to perform disposal.
Disposed system	System (product or service) that has been deactivated, disassembled, and removed

Typical Input/Output	Description
	from operations and been properly disposed.
Enabling systems	External systems that facilitate the life cycle activities of the SoI but are not a direct element of the operational environment.
Human resource management records/artifacts	Permanent, readable form of data, information, or knowledge related to human resource management.
Human resource management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the human resource management activities.
Human resource management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform human resource management.
Implementation records/artifacts	Permanent, readable form of data, information, or knowledge related to implementation.
Implementation report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the implementation activities.

Typical Input/Output	Description
Implementation strategy/approach	Approaches, schedules, resources, and specific considerations required to perform implementation.
Information management records/artifacts	Permanent, readable form of data, information, or knowledge related to information management.
Information management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the information management activities.
Information management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform information management.
Information register	A repository that supports the availability for use and communication of all relevant project information artifacts in a timely, complete, valid, and, if required, restricted manner.
Infrastructure management records/artifacts	Permanent, readable form of data, information, or knowledge related to infrastructure management.
Infrastructure management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the infrastructure management activities.

Typical Input/Output	Description
Infrastructure management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform infrastructure management.
Installation procedure	Presents an ordered series of steps to perform transition.
Installed system	System (product or service) that has been installed in its operational environment.
Integrated system or system element	System or system element (product or service) that has been aggregated from system elements.
Integration procedure	Presents an ordered series of steps to perform integration.
Integration records/artifacts	Permanent, readable form of data, information, or knowledge related to integration.
Integration report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the integration activities.
Integration strategy/approach	Approaches, schedules, resources, and specific considerations required to perform integration.
Knowledge management records/artifacts	Permanent, readable form of data, information, or knowledge related to

Typical Input/Output	Description
	knowledge management.
Knowledge management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the knowledge management activities.
Knowledge management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform knowledge management.
Knowledge management system	System used to support and enable knowledge management.
Life cycle concepts	Articulation and refinement of the various life cycle concepts consistent with the stakeholder needs. Typical concepts include: acquisition concept; deployment concept; operational concept (OpsCon); support concept; retirement concept.
Life cycle model management records/artifacts	Permanent, readable form of data, information, or knowledge related to life cycle model management.
Life cycle model management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the life cycle model management activities.
Life cycle model management	Approaches, schedules, resources, and specific considerations required to

Typical Input/Output	Description
strategy/approach	perform life cycle model management.
Life cycle models	Framework of processes and activities concerned with the life cycle that can be organized into stages, acting as a common reference for communication and understanding. (ISO/IEC/IEEE 15288, 2023)
Maintained and sustained system	System (product or service) that has been maintained for use in its operational environment.
Maintenance and logistics procedure	Presents an ordered series of steps to perform maintenance.
Maintenance and logistics records/artifacts	Permanent, readable form of data, information, or knowledge related to maintenance.
Maintenance and logistics report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the maintenance activities.
Maintenance and logistics strategy/approach	Approaches, schedules, resources, and specific considerations required to perform maintenance.
Measurement data	Measurement data from all applicable life cycle processes, including: critical performance measurement data,

Typical Input/Output	Description
	organizational measurement data, and project measurement data.
Measurement needs	Measurement needs from all applicable life cycle processes, including: critical performance measurement needs, organizational measurement needs, and project measurement needs.
Measurement records/artifacts	Permanent, readable form of data, information, or knowledge related to measurement.
Measurement register	A repository that supports the availability for use and communication of all relevant measures in a timely, complete, valid, and, if required, confidential manner.
Measurement report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the measurement activities.
Measurement strategy/approach	Approaches, schedules, resources, and specific considerations required to perform measurement.
Operation procedure	Presents an ordered series of steps to perform operation.
Operation records/artifacts	Permanent, readable form of data, information, or knowledge related to

Typical Input/Output	Description
	operation.
Operation report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the operation activities.
Operation strategy/approach	Approaches, schedules, resources, and specific considerations required to perform operation.
Operational system	System (product or service) being used in its operational environment.
Organization infrastructure	Resources, facilities, personnel, and/or services that support the organization.
Organization infrastructure needs	Identified organizational infrastructure needs.
Organization lessons learned	Organizational-related lessons learned. Results from an evaluation or observation of an implemented corrective action that contributed to improved performance or increased capability. A lesson learned also results from an evaluation or observation of a positive finding that did not necessarily require corrective action other than sustainment.
Organization policies	High-level direction at the organizational level consistent with the organization's

Typical Input/Output	Description
	strategies. (Adapted from ISO/IEC/IEEE 15289, 2019)
Organization portfolio direction and constraints	Organization direction and constraints related to the project portfolio.
Organization procedures	Presents an ordered series of steps to perform a process, activity, or task for an organization. (Adapted from ISO/IEC/IEEE 15289, 2019)
Organization processes	Set of interrelated or interacting activities that transform inputs into outputs for an organization. (Adapted from ISO/IEC/IEEE 15288, 2023)
Organization reports	Reports from all applicable organization life cycle processes, including: life cycle model management report, infrastructure management report, portfolio management report, human resource management report, quality management report, and knowledge management report.
Organization strategic plan	The overall organization strategy, including the business mission or vision and strategic goals and objectives.
Organization strategies/approaches	Strategies/approaches for all applicable organization life processes, including: life cycle model management strategy/approach, infrastructure

Typical Input/Output	Description
	management strategy/approach, portfolio management strategy/approach, human resource management strategy/approach, quality management strategy/approach, and knowledge management strategy/approach.
Organization tailoring strategy/approach	Organization's specific strategy and approach to tailoring required to incorporate new or updated external standards.
Organizational measurement data	Data provided for the identified organizational measurement needs.
Organizational measurement needs	Identified information needs of the decision makers with respect to organizational expectations.
Other validated artifacts	Artifacts that are validated
Other verified artifacts	Artifacts that are verified
Portfolio management records/artifacts	Permanent, readable form of data, information, or knowledge related to portfolio management.
Portfolio management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the portfolio management activities.

Typical Input/Output	Description
Portfolio management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform portfolio management.
Problem or opportunity statement	Description of the problem or opportunity. Should be derived from the organization strategy and provide enough detail to understand the gap or new capability that is being considered
Project assessment and control records/artifacts	Permanent, readable form of data, information, or knowledge related to project assessment and control.
Project assessment and control strategy/approach	Approaches, schedules, resources, and specific considerations required to perform project assessment and control.
Project authorization	Authorization from the organization to proceed per the agreed-to project plan.
Project authorization request	Request from the project to the organization to authorize the project.
Project budget	Estimate of the costs associated with a particular project. Includes labor, infrastructure, acquisition, and enabling system costs along with reserves for risk management.
Project constraints	Externally imposed limitation on the project developing or modifying a

Typical Input/Output	Description
	system. (ISO/IEC/IEEE 29148, 2018)
Project control request	Project directives based on action required due to deviations from the project plan. If assessments are associated with a decision gate, a decision to proceed or not to proceed, is taken.
Project decision gate/review result	Decision gate and review artifacts that are expected through conduct of the decision gate or technical review and that can be considered elements of exit criteria. (Adapted from ISO/IEC/IEEE 24748–8, 2019)
Project direction	Organizational direction to the project. Includes sustainment of projects meeting objectives and redirection or termination of projects not meeting objectives.
Project human resource needs	Identified human resource needs of the project.
Project infrastructure	Resources, facilities, personnel, and/or services that support the project.
Project infrastructure needs	Identified infrastructure needs of the project.
Project lessons learned	Project-related lessons learned. Results from an evaluation or observation of an

Typical Input/Output	Description
	implemented corrective action that contributed to improved performance or increased capability. A lesson learned also results from an evaluation or observation of a positive finding that did not necessarily require corrective action other than sustainment.
Project measurement data	Data provided for the identified project measurement needs.
Project measurement needs	Identified information needs of the decision makers with respect to project expectations.
Project objectives	The objectives or goals for the project.
Project planning records/artifacts	Permanent, readable form of data, information, or knowledge related to project planning.
Project portfolio	Collection of projects that addresses the strategic objectives of the organization. (ISO/IEC/IEEE 12207, 2017)
Project procedures	Procedures from all applicable life cycle processes, including: integration procedure, verification procedure, installation procedure, validation procedure, operation procedure, maintenance and logistics procedure, and disposal procedure.

Typical Input/Output	Description
Project reports	Reports from all applicable project life cycle processes, including: acquisition report, supply report, decision management report, risk management report, configuration management report, configuration verification and audit report, information management report, measurement report, quality assurance report, quality assurance evaluation report, business or mission analysis report, stakeholder needs and requirements definition report, system requirements definition report, system architecture definition report, system architecture assessment report, design definition report, system design assessment report, system analysis report, implementation report, integration report, verification report, transition report, validation report, operation report, maintenance and logistics report, and disposal report.
Project schedule	A linked list of a project's milestones, activities, and deliverables with intended start and finish dates May include a top-level milestone schedule and multiple levels (also called tiers) of schedules of increasing detail and task descriptions with completion criteria and work authorizations.

Typical Input/Output	Description
Project status report/dashboard	Provides results of monitoring the execution of the defined plan or processes for internal or external distribution. It includes a summary of decisions, monitoring results, action items, process or performance data, and recorded process improvements. It assesses the degree of adherence to the plans. (Adapted from ISO/IEC/IEEE 15289, 2019)
Project strategies/approaches	Strategies/approaches for all applicable project life processes, including: acquisition strategy/approach, supply strategy & approach, project assessment and control strategy/approach, decision management strategy/approach, risk management strategy/approach, configuration management strategy/approach, information management strategy/approach, measurement strategy/approach, quality assurance strategy/approach, business or mission analysis strategy/approach, stakeholder needs and requirements definition strategy/approach, system requirements definition strategy/approach, system architecture definition strategy/approach, design definition strategy/approach, system analysis strategy/approach, implementation strategy/approach,

Typical Input/Output	Description
	integration strategy/approach, verification strategy/approach, transition strategy/approach, validation strategy/approach, operation strategy/approach, maintenance and logistics strategy/approach, and disposal strategy/approach.
Project tailoring strategy/approach	Project's specific strategy and approach to tailoring required to incorporate new or updated life cycle models.
Qualified personnel	Individuals equipped to perform duties on behalf of the organization, including officers, employees, and contractors. (Adapted from ISO/IEC/IEEE 24765, 2017)
Quality assurance corrective action	Action to eliminate the cause or reduce the likelihood of recurrence of a detected project nonconformity or other undesirable situation. (Adapted from ISO/IEC 19770–1, 2017)
Quality assurance evaluation report	Provides results of quality assurance evaluations. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
Quality assurance records/artifacts	Permanent, readable form of data, information, or knowledge related to quality assurance.

Typical Input/Output	Description
Quality assurance report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the quality assurance activities.
Quality assurance strategy/approach	Approaches, schedules, resources, and specific considerations required to perform quality assurance.
Quality assurance system	System used to support and enable quality assurance.
Quality management corrective action	Action to eliminate the cause or reduce the likelihood of recurrence of a detected organizational nonconformity or other undesirable situation. (Adapted from ISO/IEC 19770–1, 2017)
Quality management criteria and methods	Rules on which a judgment or decision can be based, or by which an organization can be evaluated. (Adapted from ISO/IEC/IEEE 15289, 2019)
Quality management evaluation report	Provides results of quality management evaluations. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
Quality management records/artifacts	Permanent, readable form of data, information, or knowledge related to quality management.

Typical Input/Output	Description
Quality management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the quality management activities.
Quality management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform quality management.
Quality management system	System used to support and enable quality management.
Records/artifacts	Records from all applicable life cycle processes, including: acquisition records/artifacts, supply records/artifacts, life cycle model management records/artifacts, infrastructure management records/artifacts, portfolio management records/artifacts, human resource management records/artifacts, quality management records/artifacts, knowledge management records/artifacts, project planning records/artifacts, project assessment and control records/artifacts, decision management records/artifacts, risk management records/artifacts, configuration management records/artifacts, information management records/artifacts, measurement records/artifacts, quality

Typical Input/Output	Description
	assurance records/artifacts, business or mission analysis records/artifacts, stakeholder needs and requirements definition records/artifacts, system requirements definition records/artifacts, system architecture definition records/artifacts, design definition records/artifacts, system analysis records/artifacts, implementation records/artifacts, integration records/artifacts, verification records/artifacts, transition records/artifacts, validation records/artifacts, operation records/artifacts, maintenance and logistics records/artifacts, disposal records/artifacts, tailoring records/artifacts.
Request for supply	Acquirer's request for information and commitments needed from the supplier that are required to be included in the potential supplier's proposal. It announces the acquirer's intention to potential bidders to acquire a specified system or system element (product or service). (Adapted from ISO/IEC/IEEE 15289, 2019)
Requirements imposed on enabling systems	Identified requirements for enabling systems of the system-of-interest.

Typical Input/Output	Description
Reused system or system element	System or system element (product or service) reused by an organization consistent with its system element requirements.
Risk management records/artifacts	Permanent, readable form of data, information, or knowledge related to risk management.
Risk management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the risk management activities.
Risk management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform risk management.
Risk register	A repository that supports the availability for use and communication of all relevant risk information in a timely, complete, valid, and, if required, confidential manner.
Source documents	External documents relevant to the particular stage of the system of interest.
Stakeholder identification	List of individuals or organizations having a right, share, claim, or interest in a system or in its possession of characteristics that meet their needs and

Typical Input/Output	Description
	expectations. (Adapted from ISO/IEC/IEEE 15288, 2023)
Stakeholder needs and requirements	Structured collection of the requirements [characteristics, context, concepts, constraints and priorities] of the stakeholder and the relationship to the external environment. (ISO/IEC/IEEE 29148, 2018)
Stakeholder needs and requirements definition records/artifacts	Permanent, readable form of data, information, or knowledge related to stakeholder needs and requirements definition.
Stakeholder needs and requirements definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the stakeholder needs and requirements definition activities.
Stakeholder needs and requirements definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform stakeholder needs and requirements definition.
Standards	This handbook and relevant industry, country, military, acquirer, and other specifications and standards. Includes new knowledge from industry sponsored knowledge networks.

Typical Input/Output	Description
Supplied system or system element	System or system element (product or service) delivered from a supplier to an acquirer consistent with the delivery conditions of the supply agreement.
Supply agreement	Mutual acknowledgment of terms and conditions under which a working relationship is conducted between a supplier and an acquirer. (Adapted from ISO/IEC/IEEE 15288, 2023)
Supply payment	Payments or other compensations for the supplied system.
Supply records/artifacts	Permanent, readable form of data, information, or knowledge related to supply.
Supply report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the supply activities.
Supply response	Prepared by a potential supplier to support the offer of a contract bid, including cost, schedule, risk statements, methodology to satisfy the request for supply, experiences and capabilities, any recommendations to tailor the request for supply or contract, and the signature of the supplier's approving authority. Informally, may be prepared within an

Typical Input/Output	Description
	organization. (Adapted from ISO/IEC/IEEE 15289, 2019)
Supply strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform supply.
System analysis records/artifacts	Permanent, readable form of data, information, or knowledge related to system analysis.
System analysis report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system analysis activities.
System analysis request	A request to conduct a system analysis.
System analysis strategy/approach	Approaches, schedules, resources, and specific considerations required to perform system analysis.
System architecture assessment report	Provides results of system architecture assessments. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
System architecture definition records/artifacts	Permanent, readable form of data, information, or knowledge related to system architecture definition.

Typical Input/Output	Description
System architecture definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system architecture definition activities.
System architecture definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform system architecture definition.
System architecture description	The fundamental conception of a system-of-interest in terms of its purpose, system qualities (such as feasibility, performance, safety, and interoperability), constraints, and design decisions and rationale. Identification of the architecture's stakeholders and the stakeholders' architecture-related concerns. (Adapted from ISO/IEC/IEEE 15289, 2019)
System architecture rationale	Rationale for architecture selection, technological/technical system element selection, and allocation between system requirements and architectural entities.
System design assessment report	Provides results of system design assessments. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
System design characteristics	Design attributes or distinguishing features that pertain to a measurable

Typical Input/Output	Description
	description of a product or service. (ISO/IEC/IEEE 15288, 2023)
System design description	Describes the design of a system or element. (Adapted from ISO/IEC/IEEE 24765, 2017)
System design rationale	Rationale for design selection, system element selection, and allocation between system requirements and system elements. Includes rationale of major selected implementation options and enablers.
System element	System element (product or service) implemented consistent with its system element requirements.
System element description	Applies the system architecture description to the low-level system configuration items and elements. It is at a level of detail to permit design, implementation, and test. (Adapted from ISO/IEC/IEEE 15289, 2019)
System interface definition	Description of the interfaces between systems and system elements. (Adapted from ISO/IEC/IEEE 24765, 2017)
System requirements	Structured collection of the requirements [functions, performance, design constraints, and other attributes] for the system and its operational

Typical Input/Output	Description
	environments and external interfaces. (ISO/IEC/IEEE 29148, 2018)
System requirements definition records/artifacts	Permanent, readable form of data, information, or knowledge related to system requirements definition.
System requirements definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system requirements definition activities.
System requirements definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform system requirements definition.
System viewpoints, views, and models	Definitions of viewpoints to document the procedures for creating, interpreting, analyzing, and evaluating architectural data. One or more views of the system. Each architectural view is a representation of the complete system from the perspective of one or more system concerns, for its stakeholders. (Adapted from ISO/IEC/IEEE 15289, 2019)

Typical Input/Output	Description
Systems engineering management plan (SEMP)	Presents how the project processes and activities are executed to assure the project's successful completion, and the quality of the deliverable product or service. (Adapted from ISO/IEC/IEEE 15289, 2019)
Tailoring records/artifacts	Permanent, readable form of data, information, or knowledge related to tailoring.
Traceability mapping	Records the relationship between two or more artifacts of the development process (e.g., requirements, functions, system elements, verifications, and validations, tasks). (Adapted from ISO/IEC/IEEE 24765, 2017)
Trained personnel	Trained individuals or organizations that perform the operation, maintenance, or other functions of or for a system.
Training materials	Materials for the provision of formal and informal learning activities. (Adapted from ISO/IEC/IEEE 24765, 2017)
Transition records/artifacts	Permanent, readable form of data, information, or knowledge related to transition.
Transition report	An account prepared for interested parties in order to communicate the